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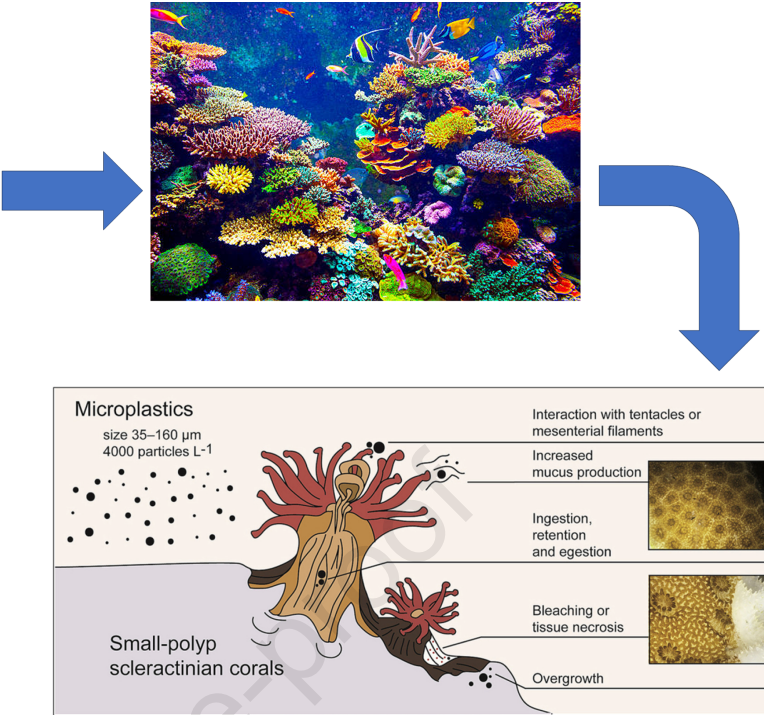
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Physiological and biochemical effects of microplastics on corals: Environmental and ecological impacts and mitigation strategies and policy options

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Abstract

Microplastic pollution is a growing concern for coral reef ecosystems. It is adversely affecting coral health, biodiversity, and overall resilience. microplastics alter coral feeding behaviour which may reduce the food intake and energy depletion. The main objective of this paper is to discuss the altered physiological and biochemical behaviors of corals, related to bleaching, tissue necrosis, reproduction, occurred due to microplastic and microplastic-associated contaminants in marine ecosystems. The ingestion of microplastics induces oxidative stress in coral tissues by triggering the excessive production of reactive oxygen species (ROS) which may weaken the coral defense mechanisms and make them more vulnerable to diseases. Reproductive success is compromised as microplastic exposure reduces larval settlement and survival rates. microplastics also act as vectors for contaminants such as heavy metals, persistent organic pollutants (POPs), and pathogenic microorganisms. Moreover, the anthozoan-algae symbiotic relationship, fundamental to coral survival and reef building, is threatened by microplastic pollution, which can trigger coral bleaching. To mitigate the impacts of microplastic pollution in coral ecosystems, it is essential to develop and to use biodegradable alternatives of traditional plastics. Additionally, large-scale ocean clean-up initiatives related to plastic waste reduction, public awareness, and the international regulations on plastic waste management are also essential steps in this regard.

Keywords: Corals, Microplastics, Oxidative stress, Physiology, Mitigation, policy options

1. Introduction

Plastic pollution is a serious global concern and a threat to the environment, affecting the health of humans and other organisms (Boucher and Friot, 2017; de Souza Machado et al., 2018; Li et al., 2018). This is attributed to increased global plastic production over the past 50 years with over

390 million tons produced annually as of 2021 (Plastics Europe, 2022). According to the United Nations Environment Programme (UNEP, 2023), the production of plastics has exponentially increased with more than 430 million tons of plastics produced annually, of which, two-thirds of them become waste and end up in the ecosystem. The disposal of these large amounts of plastic waste is a challenge, so they accumulate in the environment and mainly end up in aquatic ecosystems in the form of macroplastics or microplastics (Egbeocha et al., 2018). Microplastics refer to plastic fragments <5mm in size (Arthur et al., 2009) and are mainly classified into primary microplastics which are commercially manufactured and released in the environment from personal care products such as cleansers, toothpaste, various cosmetics and synthetic textiles, air blasting, wear and tear of vehicular tires, and numerous cleaning products (Auta et al., 2017). The secondary microplastics are arising from the degradation or fragmentation of larger plastic particles (Cole et al. 2011; Hale et al., 2020; Wang et al., 2021). Microplastic particles are reported from coral ecosystems and from coral bodies from different parts of the world including Hawaii islands (Axworthy et al., 2024), Sri Lanka (Hansani et al., 2023), Great barrier Reef Australia (Hall et al., 2015; Jensen et al., 2019), Maldives (Saliu et al., 2019, 2018p; Patti et al., 2020), French Polynesia (Connors, 2017), South China Sea (Ding et al., 2019; Wang et al., 2019; Huang et al., 2019), Gulf of Mannar (Patterson et al., 2020), Hong Kong (Cheang et al., 2018), Caribbean Sea (Portz et al., 2020), Red Sea (Marti et al., 2017), Rhode Island (Rotjan et al., 2019), East Mingulay Special area, Scotland (La Beur et al., 2019). microplastics can also act as a source and a sink for heavy metals and organic contaminants (Rochman et al., 2014; Teuten et al., 2007; 2009), which also pose a high toxicity threat to organisms. According to a review by De Sá, (2018), exposure to microplastics have a variety of physiological, behavioral, and ecotoxicological consequences on aquatic species. Reduced feeding

activity (Bakir et al., 2016), growth suppression, genotoxicity (Prokić et al., 2019), and decreased reproductive fitness (Besseling et al., 2014) are a few of these impacts. In contrast, some research (Hankins et al., 2018; Santana et al., 2018) indicated that exposure to microplastics had only little impact. However, a study by Arellano et al. (2025) showed that natural water extract from the medicinal plant *Panax notoginseng* (at 50 and 250 µg/mL) can reduce heightened inflammation in macrophage immune cells and in shrimps (*Litopenaeus vannamei*) showed better balance, stronger immune responses, higher phagocytosis (ability to fight germs), increased respiratory burst, phenol oxidase activity, and better antioxidant protection through SOD enzyme upon exposure to Polystyrene nanoplastic.

Coral reef ecosystems are among the most complex and highly productive marine ecosystems with high biodiversity and provide a range of ecosystem services including coastal protection, supporting biodiversity, maintaining marine economy and food security among the coastal population (Patterson et al., 2020; Fisher et al., 2015). Scleractinian corals are one of the fundamental components of coral reef ecosystems and serve a crucial role in providing vital structural support, food, and shelter for an extensive range of associated organisms. A proportion of about one-third of all marine species are found in coral reefs (Plaisance et al., 2011) but sadly, one-third of the coral species that build these reefs are facing a high risk of extinction (Carpenter et al., 2008). Coral reefs are sensitive ecosystems that are readily damaged by environmental stresses, such as those brought on by climatic changes, global temperature fluctuations, ocean acidification, pollution, urbanization, fishing activities, oil spills, marine litter, mining, etc. (John et al., 2020; US-EPA 2025). Natural and anthropogenic release of pollutants into the environment has continued to threaten coral reef ecosystems due to enormous contamination exposure. Microplastic pollution in particular has a high abundance in the reef environment (Zhou et al.,

2022; Pantos, 2022; Bian et al., 2024). Different coral species have different morphologies, feeding patterns, and particle clearance processes, which affect how they respond to microplastic contamination (Reichert et al., 2019; Reichert et al., 2018). Corals exposed to microplastic contamination suffer from many toxic effects including subnormal fertilization, reduced skeletal growth, and defective algal symbiosis (Bielmyer et al., 2018; Berry et al., 2019).

The majority of research on microplastics has been in shallow-water settings, despite the fact that the deep sea would be a major sink for plastic wastes (Chapron et al., 2018). Less information is available concerning the interaction of microplastics with biota in deep-sea reefs built by cold-water corals (also called deep-sea corals) (Chapron et al., 2025). Cold-water coral ecosystems tend to accumulate plastics due to the structural complexity of these coral reefs (D'onghia et al., 2017), and it has been discovered that at least some deep-sea coral species ingest microplastics (Taylor et al., 2016; Hankins et al., 2023; Bian et al., 2024). In contrast to their shallow-water counterparts, deep-water corals are entirely heterotrophic since they lack Symbiodiniaceae symbionts, making them potentially vulnerable to all the impacts that occur after adhesion and ingestion of microplastics (de Oliveira Soares et al., 2020). The negative impacts of microplastics on these corals indirectly affect a wide range of other organisms, including decapods and fish found in reef structures that serve as their habitats and breeding sites (Freiwald et al., 2004; Roberts et al., 2006). So, there is a limited information available on microplastic interaction and possible mitigation to deep coral reefs.

However, newer methods are coming up for microplastic degradation, a study by Nguyen et al. (2024) demonstrated breakdown of PE microplastics using advanced oxidation processes that combines potassium permanganate (KMnO_4 , a strong chemical oxidant) with UV light. They

tested different combination of UV light (UVA, UVB, and UVC) and showed that the KMnO_4 + UVC combination to be most effective in degradation (7.5%).

In this paper, we have developed a critical review on the multidimensional effects of microplastics on the physiology, growth, reproduction and overall ecological health in corals. Although some previous articles highlighted the effects of microplastics on corals, however, this paper for the first time covers the effects in physiological, cellular, and biochemical levels. Physiological and biochemical effects such as coral bleaching, tissue necrosis, reproduction due to both microplastic and microplastic-associated contaminants in corals are highlighted. The effects of microplastic on Anthozoan-algae symbiotic relationships are also discussed. microplastics transferred in different trophic levels in the coral ecosystem are leading to microplastic bioaccumulation. Organizing this bottom-up approach provides key benefits in building a clear conceptual hierarchy that logically progresses from direct, localized effects (e.g., ingestion or oxidative stress) to broader cumulative consequences (e.g., disease spread or food web disruptions). This method also establishes foundational mechanisms and shows how micro-scale changes cascade into larger widespread impacts. This paper, for the first time, also discusses multidimensional environmental impacts of microplastic pollution on both marine and freshwater corals with their possible mitigation options.

2. Sources of microplastic pollution in freshwater and marine systems

It is evident that microplastic originate from various sources such as plastic waste disposal, industrial processes, and personal care products; and widespread in the atmosphere, soil, surface water, and groundwater which often find their way into marine ecosystems. In a worldwide study by Browne et al., (2011), in which global samples were obtained from different shorelines, it was found that eighteen sites across all six continents were contaminated by microplastics with abundance in samples containing polyester (56%), acrylic (23%), polypropylene (7%),

polyethylene (6%), and polyamide fibers (3%). Interestingly, there was more microplastic in areas with high human population density.

Microplastics are added into the ecosystems in significant amount via secondary sources, such as mechanical breakdown, UV light exposure that break larger plastics into smaller and finer parts. Some amount of microplastics is deposited in sediments, whereas the rest amount remains suspended on the water surface, from where marine organisms can ingest the particles. Because of their sizes, microplastics can escape the filtering systems of wastewater treatment plants, and subsequently can enter rivers and sea ecosystems.

The awareness of the occurrence and impacts of plastics in marine ecosystems started with many reports during the 1970s where plastic particles were detected in the corpses of sea birds, in their intestines, and in the regurgitated food pellets by sea birds (Parslow and Jefferies, 1972; Rothstein, 1973; Hays and Cormons, 1974; Baltz and Morejohn, 1976; Ohlendorf et al., 1978). However, research on the impacts of microplastics on marine systems at the ecosystem level is still limited (Pauna et al., 2019) despite the exponential increase in marine microplastic pollution research in the last decades as a result of increased global plastic production (Brandon et al., 2019). Notably, over 90% of the debris in marine litter is plastics (Galgani et al., 2015; UN-EP, 2021).

The sources of microplastics in marine ecosystems are either primary or secondary. Effluents from wastewater treatment plants (WWTPs) are another source of microplastics in the water bodies (Barchiesi et al., 2020). Microplastics contained in cosmetics and personal care products which eventually end up in seas and oceans (Juliano and Magrini, 2017; Fendall, and Sewell, 2009). Domestic effluents and runoff from agricultural fields also contributes to microplastic contamination in water bodies (Bläsing and Amelung, 2018; Hurley and Nizzetto, 2018; Kim et

al., 2021). Physical forces in the form of strong winds and wave action can also carry microplastics into the water bodies (Andrady, 2011).

Another pathway for entry of microplastics into ocean involves the degradation of large plastics originating from various sources for instance fibers from fishing equipment, shipping activities, plastic waste from offshore recreation, and industrial activities (Lusher et al., 2015). When large plastics are exposed to certain environmental factors like water, high temperature, sunlight, and microbes they undergo degradation by hydrolysis, thermal degradation, photodegradation, and biodegradation respectively to release secondary microplastics in water. However, light-induced oxidation/photo-oxidative degradation of plastics is predominant and hydrolysis action is not of significance in seawater (Andrady et al., 2022; Dimassi et al., 2022).

Microplastics are now frequently being reported from both onshore and offshore reef regions, which are mostly contributed by various anthropogenic causes such as riverine inputs from coastal cities, intensive fisheries and also by winds (Critchell et al., 2019; Jensen et al., 2019; Huang et al., 2019; Kumar et al., 2022). Microplastics can be transported and transferred from oceans to beaches and coastal areas and vice-versa (Schwarz et al., 2019). Tourism and other recreational activities also play important roles in accumulation of microplastics in the reef environments. Even activities such as land reclamation can act as drivers for microplastic accumulation. Also, atmospheric long-distance transport might play a significant role (Ryan et al., 2009). A study by Chen et al. (2023) investigated microplastic amount in the Taiwanese coral reef sand by normal organic matter digestion using H_2O_2 and another digestion using HCl and found twice the amount of microplastic particles (38 Vs 78 particles/Kg sand). They concluded that historically microplastic abundance in the coral reefs is significantly underscored since much of them remains

hidden inside tiny mineral pieces like coral fragments and shells and coral reefs are more polluted than previously thought.

Figure 1 shows the global abundance of microplastics in coral reef ecosystems.

3. Mode of Ingestion of microplastics by corals

Many marine organisms may ingest microplastics (Egbeocha et al., 2018). Similarity of microplastic particles between corals and reef environments indicates that corals might have enriched microplastics from water and sediments (Hansani et al., 2023). Reports show that corals can ingest plastic fragments, fibers, and microbeads ranging in size from 3 to 5000 μm , and they can retain ingested plastics for over 24 hours (Allen et al., 2017; Hall et al., 2015; Hankins et al., 2018; Martin et al., 2019; Pantos, 2022). *Scleractinian* corals (*M. cavernosa* and *O. faveolata*) ingest microplastics (425 μm to 2.8 mm) but within 48 hrs most of the microplastics are egested, however small-sized Microplastics (212–250 μm) did not elicit any feeding response (Hankins et al., 2018). Ingestion of microplastics present in the water by corals occurs either directly or indirectly. The direct mode of ingestion of microplastics occurs when the corals take up the fragments directly through the mouth opening into their gastrovascular cavity during feeding. This is explained by the fact that the presence of microplastics can be mistaken for their food (Hall et al., 2015; Martin et al., 2022) due to their different colors, sizes, and shapes, and these properties determine the type of microplastics taken up by corals, for example, most corals prefer microplastics that are transparent, smaller than 2mm in size and film shaped. However, this could be species-specific (Zhou et al., 2022). The structures of coral colonies have a huge effect on trapping of microplastic particles. A study by Yen et al. (2024) revealed that coral colonies having compact structure rather than open structured ones trapped higher numbers of microplastic particles, however their surface morphology has no significant effect. This study highlights the

effectiveness of corals in trapping microplastic particles and that some particular types of corals are potential at much higher risks than others. Similar results were advocated by Mendrik et al. (2024), who reported microplastic trapping efficiencies for higher canopy densities in the range 79-98% compared to bare bed and 15% more microplastic trapping compared to less dense canopies. Microplastics were mainly trapped through particle inception, settlement of coral surface and accumulation in the downstream region. The microplastics were mainly retained through the energy-dissipative effect of corals due to flow structure interactions reducing in-canopy velocities and generating localized turbulence (Mendrik et al., 2024).

Corals obtain their food from autotrophic sources by forming endosymbiotic associations with the photosynthetic algae called zooxanthellae (Houlbrèque and Ferrier-Pagès 2009), however, corals can also feed heterotrophically on planktonic organisms as their prey (Hankins et al., 2018). This provides the indirect mode of ingestion of microplastics in which the corals, by feeding on certain prey species, end up consuming these contaminants contained in the prey (Rotjan et al., 2019). Most of the common prey of corals including amphipods, copepods, and other zooplanktons ingest microplastics (Ferrier-Pagès et al. 2003).

In addition to active ingestion of microplastics, corals can strongly attach the particles on their body surface by adhesion which is the passive means (Corona et al., 2020; Ding et al., 2019; Pantos, 2022). Interestingly, corals can remove microplastics from water by adhesion at rates 40 times higher than by ingestion hence adhesion is more effective (Martin et al., 2019). This also makes corals a significant sink for microplastics (Reichert et al., 2022). In the study by Martin et al. (2019), relatively low concentrations of microplastics were discovered in areas with corals, which suggested that corals can simultaneously reduce the microplastics from water. High concentration of microplastics particles (Polyethylene (PE), polyvinyl

chloride (PVC), polypropylene (PP), poly(methyl methacrylate) (PMMA), Rayon, and Nylon particles) were found to be ingested and sticking on the body surface of four coral species *Favia gravida*, *Mussismilia hispida*, *Montastrea cavernosa*, and *Siderastrea stellata* (Rani-Borges et al., 2023).

Figure 2 shows attachment and subsequent ingestion of microplastics by corals, leading to physiological effects on corals.

4. Impacts of microplastics on the feeding behavior of corals

Research has unequivocally demonstrated the profound impact of microplastics on the feeding patterns of corals (Corinaldesi et al., 2021). The initial influence of microplastics on coral feeding behavior appears to be stimulatory, prompting corals to consume more non-food particles, particularly plastic polymers. This behavioral shift is rooted in the reliance of corals on their sense of smell, specifically through chemoreceptors, for prey capture. Compounds present in microplastics act as phagostimulants, essentially tricking the corals into perceiving them as real food. As a result, the corals may mistakenly consume microplastics for actual nutritional sources, leading to increased ingestion (Allen et al., 2017; Hall et al., 2015).

The consequences of this altered feeding pattern extend beyond the immediate ingestion of non-nutritive particles. Over time, the continuous consumption of microplastics poses a threat to the health and well-being of corals (Pantos, 2022). Occasionally species-specific feeding effects were also reported such as, *P. verrucosa* is more susceptible to microplastics contamination than other coral because of structural obstruction of its feeding processes (Reichert et al., 2019). *Acropora* sp. is substantially more sensitive to microplastics than *Seriatopora hystrix* (Mendrik et al., 2021). The accumulation of plastic polymers within the coral tissues can disrupt vital physiological processes and compromise the overall resilience of coral ecosystems. The intricate interplay

between microplastics and coral feeding underscores the urgent need for comprehensive strategies to mitigate plastic pollution in marine environments to safeguard the delicate balance of coral ecosystems (Corinaldesi et al., 2021; Allen et al., 2017).

To address this environmental challenge effectively, interdisciplinary research efforts are imperative. Collaborative initiatives involving marine biologists, ecologists, chemists, and co-developing policy options with policymakers are essential to develop holistic solutions that encompass the entire life cycle of plastics, from production to disposal. Implementing such multidimensional approaches will not only help preserve the intricate beauty of coral reefs but also contribute to the broader conservation of marine biodiversity (Corinaldesi et al., 2021; Hall et al., 2015).

5. Impacts of microplastics on physiological and biochemical processes in corals

Ingestion of microplastics by corals can lead to various negative impacts, including digestive tract irritation, reduced feeding efficiency, and increased susceptibility to diseases (Cole et al., 2013). Furthermore, microplastics can act as carriers for harmful contaminants, exacerbating their toxic effects on corals. Exposure to low concentrations of microplastics in corals for a short time period results in short-term impacts on the physiological processes. Still, there is a possibility of long-term effects that can result when corals are exposed to low concentrations of microplastics over a long period of time (Liao et al., 2021). However, the impacts can vary and are species-specific (Reichert et al., 2019), and are also determined by the exposure time with corals exposed for longer durations to microplastic fragments experiencing more adverse impacts (Lancôt et al., 2020). Importantly, corals are known to be more vulnerable to contaminants and microplastic particles in water during their early growth stages due to their small size and weak defensive system (Fabricius, 2005; Jones et al., 2015). For example, coral embryos are particularly vulnerable to physical

274 damage by fragments because they lack a protective mucus barrier (Heyward and Negri, 2012).
 275 The size of the microplastics ingested is another critical factor that determines the degree of the
 276 toxicological effect suffered by marine organisms (Sharifinia et al., 2020).
 277 Ingestion of microplastics (PVC, PE, PET etc.) by *Tubastrea aurea* inhibit antioxidant capacity,
 278 immune system, calcification and energy metabolism of the coral. Specifically, PVC reduces
 279 Superoxide dismutase (SOD), catalase (CAT), alkaline phosphatase (AKP), and total antioxidant
 280 capacity (TAC) by 29.4%, 35.5%, 73.9%, and 52.2% respectively (Liao et al., 2021). While, PET
 281 microplastics reduced enzymatic activity of pyruvate kinase (PK), Na, K-ATPase (Na, K-ATP),
 282 Ca-ATPase (Ca-ATP), Mg-ATPase (Mg-ATP), Ca-Mg-ATPase (Ca, Mg-ATP), and glutathione
 283 (GSH) by 89.6%, 66.7%, 63.6%, 60.4%, 48.4%, and 50.5% respectively (Liao et al., 2021).
 284 *Scleractinian* coral *Stylophora pistillata* showed reduced photosynthetic efficiency of photosystem
 285 II upon exposure to microplastic (50 particles/mL concentration). Both moderate and high (5 and
 286 50 particles/mL) concentrations of microplastics caused subtle but significant alterations to
 287 metabolite profiles of coral. The exposed corals have high levels of phosphorylated sugars and
 288 pyrimidine nucleobases that make up nucleotides, scyllo-inositol and a region containing
 289 overlapping proline and glutamate signals, compared to control animals (Lanctôt et al., 2020). In
 290 South China Sea, coral species namely *Pocillopora* exhibited altered photosynthetic capacity and
 291 immune responses, with surface-adhered microplastics (>500 µm PET fibers) outnumbering
 292 internal ones (20–200 µm PE fragments) (Lin et al., 2024). Immune-agitative states, including
 293 inflammation and pathogen adhesion, further weaken corals, as microplastics act as vectors for
 294 toxins, heavy metals, and microbes. The coral *Turbinaria mesenterina* upon exposed to a solution
 295 containing polystyrene microplastic (10 µg/L) showed considerable accumulation of microplastic
 296 and immune activation like increased membrane lipid levels with 20:4 and 22:6 fatty acid chains

reallocated from the triacylglycerol pool were observed in coral host cells and symbiotic algae, respectively, which could upregulate immune activity and realign symbiotic communication in coral. High levels of polyunsaturation can sensitize the coral cell membrane to lipid peroxidation and increase cell death, which is of greater concern; additionally, the photoprotective capacity of symbiotic algae was compromised (Tang et al., 2024). Another study by Bove et al. (2023) investigated the combined effects of ocean warming, acidification, and microplastics on gene activity and microbial communities in the endangered coral *Acropora cervicornis*. The results showed that the strongest responses occurred when corals faced all stressors at once, which triggered the genes responsible for the coral's immune response.

5.1. Growth

The growth rate of corals is decreased when they are exposed to microplastics. Reichert et al., (2019) carried out an analysis of 3D scans and buoyant weight to study growth parameters, such as changes in surface area, volume, and calcification rate in hermatypic corals exposed to microplastics. The results showed a significant decrease in growth of reef-building coral, *Heliopora coerulea*, exposed to high-density polyethylene (HDPE) fragments (in size range 65-410 μm), but no significant changes were observed in three other species in the study (Reichert et al., 2019). In another study by Hankins et al., (2021) on the impacts of microplastics on the growth of two Atlantic Scleractinian species, *Pseudodiploria clivosa* and *Acropora cervicornis*, long-term exposure of corals to microplastics caused adverse effects on growth. Other studies that looked at how microplastics affected the growth rates of different coral species but some species observed significant growth reductions while others only experienced minor effects (Chapron et al., 2018; Mouchi et al., 2019). Additionally, PE microplastics are reported to weaken skeletal integrity in soft corals *Briareum violacea* upon exposure to concentrations as low as 10 mg/L also making

them more soluble (approximately 7.4 times faster). The hard coral skeleton made of strong, organized aragonite crystals turned into a weaker, messy form called amorphous calcium carbonate only after seven days of exposure (Lin et al., 2025). Corals are reported to show species-specific and microplastic-specific response, a study by Reichert et al. (2024) on two coral species *Pocillopora verrucosa* and *Stylophora pistillata* demonstrated that fibers and tire wear has the strongest effect on coral physiology. Both species demonstrated increased volume growth and photosynthetic efficiency of symbionts, but *P. verrucosa* showed a decrease in calcification rate. Overall, *P. verrucosa* responded significantly to microdebris than *S. pistillata* which could be due to their feeding behaviors. Again, a study by Tirpitz et al. (2025) on the same coral species exposed to different concentrations of microplastic (0, 0.1, 1, 10, and 100 mg·L⁻¹) for 12 weeks. The study exhibited reduced growth rates, necrosis, lower polyp activity, and an upregulation of photosynthesis, which intensified with increasing microplastic concentrations in both species. While the effects on the coral host mostly showed basic linear or nonlinear dose-response relationships, the effects on the photosymbionts revealed more complex nonlinear dose-response relationships, and photosynthesis was only upregulated after a species-specific threshold. We found that high and extreme pollution scenarios caused strong adverse effects on coral physiology, while current low to moderate concentrations had minor effects, suggesting that corals are likely to adjust with current average pollution levels (Tirpitz et al., 2025).

The negative impact on growth is explained mainly by the decreased energy reserves available for growth. microplastic ingestion is an active process, but according to a study by Hankins et al., (2018), corals expel most of the microplastic they consume. The energy invested in ingestion and the egestion of microplastics (Rotjan et al., 2019) reduces the energy available for the growth of the organism. A similar explanation is given in an experiment by Watts et al., (2015) on a different

organism Crab *Carcinus maenas* that was exposed to polypropylene rope microfibers. Another reason is the reduced food intake in the presence of plastic fragments. Ingestion of microplastics in preference to food (Hall et al., 2015) gives corals a false sense of food satisfaction thereby reducing the intake of nutritious food by corals which in turn reduces the rate of their tissue growth (Bednarz et al., 2021; Rotjan et al., 2019). The presence of microplastic particles also blocks the digestive cavity of corals, preventing food digestion (Allen et al., 2017). This reduces the respiratory energy available for growth in coral tissues.

Microplastics consumption by corals can cause severe tissue damage and also species-specific immunity impairment (Hierl et al., 2021), reduced calcification and skeletal development rate (Reichert et al., 2021), decreased anti-oxidant ability, resistant function, calcification, and calorie consumption of the coral species *Tubastrea aurea* (Liao et al., 2021). Corals exposed to microplastic contamination suffer from many toxic effects such as subnormal fertilization, reduced skeletal growth and defective algal symbiosis (Bielmyer et al., 2018).

5.2.Oxidative stress and other stress responses

Corals experience oxidative stress as a result of environmental stressors such as rising temperatures, nutrient influx caused by eutrophication, and the effects of ultraviolet (UV) radiation. Surprisingly, it has been hypothesized that microplastics may act as yet another environmental component that can affect and control corals' physiological processes resulting in oxidative stress (Tang et al., 2018; Hsieh et al., 2026). The presence of microplastics induces an excessive production of reactive oxygen species (ROS) such as superoxide radical, hydroxyl radical, and hydrogen peroxide, causing oxidative stress in the body, which in turn causes the lipid peroxidation of cell membranes and other biological damages to tissues (Mendrik et al., 2021; Liu et al., 2024). It is expected that the organisms produce more ROS after being exposed to

microplastics. Oxidative damage to tissues results from this rise in ROS (Bhattacharya et al., 2010; Chen et al., 2017; Paul-Pont et al., 2016). Furthermore, this is explained by the fact that certain antioxidant enzymes particularly superoxide dismutase (SOD), and catalase, which are normally produced in response to oxidative damage to neutralize the effects of ROS, have been detected in several coral species (Tarrant et al., 2014), hence acting as bioindicators of the occurrence of oxidative stress in corals. Oxidative stress can also produce effects at the molecular level by causing damage to cellular DNA. Microplastics-induced ROS can trigger DNA fragmentation and damage (Das, 2023; Kadac-Czapska et al., 2024). There are two different ways in which microplastics might cause genotoxicity. It may happen either as a direct result of the interaction between DNA and microplastics or indirectly by the excessive free radicals produced and oxidative stress (Manke et al. 2013).

Microfiber exposure can cause additional stress responses in corals that further impede their ability to function normally. Microfiber interactions with corals are probably taking place, though at low levels, given that they are among the most common kinds of microplastics in marine ecosystems (Browne et al., 2011; Thompson et al., 2004; Wright et al., 2013). Initial interactions with microfibers may make heterotrophic feeding more difficult. As mentioned earlier, microplastics prevent coral from consuming its normal diet because they are continuously ingested and excreted, which limits their nutrient uptake (Chapron et al., 2018; Reichert et al., 2019). Furthermore, it can be difficult for coral polyps to remove fibers from their body because they are caught and entangled in mesentery tissues (Allen et al., 2017; Hall et al., 2015). In addition, compared to other kinds of microplastics namely microspheres, microfiber entanglement with other fibers, particles, or tissue can make their removal even more difficult. Since mesentery tissues are primarily responsible for digestion, ingesting microfiber microplastics might cause clogging and feeding difficulties, which

can further disturb the regular digestion of natural food (Goldberg, 2002; Rahman et al., 2023). Microplastics also have the potential to get caught in the coral's mucus layer, where they may set off stress reactions by triggering an increase in mucus production (Hall et al., 2015; Reichert et al., 2018). The increased mucus production requires a lot of energy, which can use as much as 50% of the carbon obtained from photosynthesis (Wild et al., 2004).

5.3. Health

5.3.1. Coral diseases

Plastic waste is damaging coral health and the fragile ecosystem balance of reef-building corals making them potentially susceptible to fatal diseases (Kirstein et al., 2016). They become more susceptible to diseases due to microplastics in the environment around them. The corals' natural defenses are somehow weakened by plastic debris in different ways. For example, the obvious direct impact is physical injury caused by microplastics. This is because the adhesion of microplastics and their build-up on the coral skeleton's surface, result in frictional damage by abrasion which makes corals more vulnerable to pathogen invasion including bacteria or other elements that could cause diseases (Chen et al., 2017; Reichert et al., 2018; Lamb et al., 2018). Microplastics ingestion may cause alteration of expression levels of key genes including *MnSOD*, which serves as the first defence against ROS (Corinaldesi et al., 2021).

Microplastics act as potential vectors of highly pathogenic microorganisms such as bacteria that colonize the surfaces of these fragments (Kirstein et al., 2016) and also coral albinism pathogens (Feng et al., 2020) increasing infection risks. Reefs contaminated with plastics are 22 times more at risk compared to reefs that are highly preserved, with disease risk enhanced from 4% to 89% (Lamb et al., 2018). Plastic waste in the ocean makes reef-building corals highly susceptible to potentially fatal diseases (Kirstein et al., 2016). Masó et al., (2003) made the first observation of

the occurrence of potentially harmful microbes adhering to plastic waste. However, the word “plastisphere”, which highlights the potential for marine microplastics to act as ecological habitats for distinctive microbial communities on their surfaces, was first coined in a study by Zettler et al., (2013). This approach helped to clarify the hypothesis that different microorganisms may cling to and grow on the exteriors of microplastics in the ocean. The hypothesis that plastic materials rapidly form a thin conditioning film in seawater and act as a surface for the emergence of a biofilm made up of various bacteria has received strong support from later studies. For instance, different *Vibrio* spp. including *V. Coralliilyticus*, *V. harveyi*, *V. splendidus*, *V. parahaemolyticus*, *V. alginolyticus* and *V. fluvialis* were present in biofilms on the surfaces of common microplastics of polyethylene, polypropylene, and polystyrene synthetic polymers in water samples obtained from marine waters (Dussud et al., 2018; Kirstein et al., 2016).

An important factor is that microplastics are poorly biodegradable particles. Therefore, they act as long-term reservoirs of microbes that are causative agents of various coral diseases (Bowley et al., 2021). The most common coral diseases include White Plague which is caused by the *Aurantimonas* genus (Denner et al. 2003). Similar to this is Whitepox which has been connected to *Serratia marcescens* (Patterson et al. 2002). The *Vibrio* genus has a variety of species that have been linked to various coral diseases. *Vibrio* species, for instance, have been connected to yellow blotch syndrome (Cervino et al. 2004), white band sickness (Ritchie & Smith 1995), black band disease (Antonius, 1973), and coral bleaching (Kushmaro et al. 1996). All these diseases end up destroying coral tissues and can lead to high mortality (Richardson, 1998). These discoveries advance our knowledge of the complex relationships between corals and their environment and offer insightful information about the microbiological origins of coral diseases (Weil et al., 2006).

5.3.2. Coral bleaching

Reichert et al., (2018) observed coral bleaching and tissue necrosis in all five out of the six experimental coral species exposed to microplastics (*Acropora humilis*, *Acropora millepora*, *Pocillopora verrucosa*, *Pocillopora damicornis*, *Porites lutea*, and *Porites cylindrica*), with severe necrosis in *Pocillopora verrucosa*. In this study, high amounts of microplastic fragments were attached to the parts with necrotic tissues and the bleached parts, which provides potential evidence that microplastics contribute to these tissue damages. A study by Syakti et al., (2019) on staghorn coral (*Acropora formosa*) also revealed that corals suffer both coral bleaching and tissue necrosis when exposed to microplastics. Microplastics can exacerbate or aggravate heat tolerance in corals, with occasional protective effects but overall heightened vulnerability to bleaching during marine heatwaves (Reichert et al., 2021).

Coral bleaching refers to a condition in which corals lose their vibrant colors and turn white which occurs when *Symbiodinium*, also known as zooxanthellae, a type of photosynthetic algae that resides within the coral tissues, is partially or completely lost (Douglas, 2003). The phrase “coral holobiont” refers to this coral host and the symbiotic zooxanthellae algae association (van Oppen, and Burghardt, 2009). *Symbiodinium* contains chlorophylls a and c₂, as well as a number of carotenoids, particularly peridinin, which are responsible for the brownish appearance of corals. (Douglas, 2003). The presence of microplastics causes a significant decrease in the photosynthetic efficiency of the algal component. This is because of the physiological stress caused due to overproduction of ROS, disrupting the activity of the photosystem II, hence reducing carbon dioxide fixation (Mendrik et al., 2021). This results in the breakdown of the relationship between the coral host and its symbiotic partners. Coral bleaching and the potential death of corals result from this breakdown in the interaction.

5.3.3. Tissue necrosis

Tissue necrosis involves coral tissue breakdown and death (Hodgson, 1990). This causes obvious tissue deterioration and loss which impairs the coral organism's general health. The microplastic-induced tissue necrosis can be caused due to the entry of infectious pathogens carried in biofilms on surfaces of ingested microplastics. For example, based on the findings by Luna et al., (2007), rapid tissue necrosis in stony corals is linked to the presence of the bacteria *Vibrio harveyi*. This gives an insight into the bacterial involvement in coral-tissue necrosis. Tissue necrosis can also be caused as a result of ROS produced during microplastic-induced oxidative stress. ROS can interact with biological molecules in coral cells including DNA, proteins, and lipids. The progressive DNA damage, protein denaturation, and lipid peroxidation cause degenerative changes to occur at the cellular level, which can lead to cell dysfunction, cell senescence, cell death, and eventually decreased organism performance (Das, 2023).

5.4. Reproduction

Reproduction is a very energy-intensive activity in coral reefs, therefore, an organism's capacity to reproduce could be impacted by inadequate food intake (Sharifinia et al., 2020). As noted earlier, the presence of microplastics alters the feeding behavior of corals in which they tend to ingest these fragments in preference of actual food (Hall et al., 2015). Upon ingestion, these create a false sense of satiation (Bednarz et al., 2021; Rotjan et al., 2019). This restricts the energy available in their body which reduces the rate of reproduction. According to Sussarellu et al. (2016), the toxicity of microplastics on reproduction includes negative impacts at any point in the marine species' reproductive cycle, such as gamete formation, gamete and oocyte quality, fecundity, egg production, and sperm swimming speed. However, little is known about the ecotoxicological effects of microplastics on the reproduction of marine organisms.

Berry et al., (2019) investigated the possible impacts of two different microplastic types on the corals' three earliest life stages: fertilization, early embryo development, and larval settlement. The results showed that microplastics cause a decreased rate of fertilization, however with limited effects on embryo development and larval settlement. Microplastics particles have an impact on coral fertilization through their associations with sperms that cause sperms to sink, sperm restriction at the surface near the buoyant eggs by physically blocking the interaction between sperms and eggs, and eventually lowering the fertilization rate (Ricardo et al., 2016). Sperm restriction to eggs is also strongly influenced by particle size and type, with smaller particles with a high organic carbon content having a far more significant effect than bigger carbonate particles, which only have an impact on fertilization at extremely high concentrations (Ricardo et al., 2018).

6. Effects of microplastics-associated contaminants in corals

Microplastics are a direct threat in and of themselves, but they also act as carriers for a variety of chemical contaminants, such as heavy metals, persistent organic pollutants, and other plastic additives, and transport them to long distance (Brennecke et al., 2016; Godoy et al., 2019; Wang et al., 2018), which may cause toxic reactions in exposed organisms. Additionally, microplastics can also transport pathogenic microbes having harmful impacts on coral ecosystems and can endanger coral ecology (Hazeem et al., 2020). Recently, Patterson et al. (2020) found higher association of heavy metals with microplastics than heavy metals in surrounding sediments. Microplastics can serve as sinks and sources of environmental chemical pollutants (Li et al., 2016; Liu et al., 2019) and their transport within the coral body causes more problems (Reichert et al., 2019; Jiang et al., 2020), potentially posing a threat to coral health. The extent to which microplastics adsorb these additional environmental pollutants from their surroundings depends on several factors such as the physical and chemical properties of the microplastics (including their

rate of degradation), the specific properties of the chemical pollutant, temperature, and the surrounding water chemistry (Verla et al., 2019; Wang et al., 2018).

Following ingestion and absorption by corals, these co-occurring pollutants may either exert their toxic effects on the coral organism or accumulate in the tissues and be transferred to higher order in the food chain (Bakir et al., 2012; Keshavarzifard et al., 2017, 2020; Rochman et al., 2013; Wang et al., 2018). These co-occurring chemical pollutants include bisphenol A, polychlorinated biphenyls (PCBs), polybrominated diphenyl ethers (PBDEs), and polycyclic aromatic hydrocarbons (PAHs), and various heavy metals, such as Copper (Cu), Nickel (Ni), Cadmium (Cd), Chromium (Cr), Aluminium (Al), Lead (Pb), Zinc (Zn), Manganese (Mn), and Iron (Fe), which were found in substantial amounts in microplastic samples collected for different studies (Massos, and Turner, 2017). Aminot et al. (2020) reported that leachate from PS foam debris affected coral physiological processes such as photosynthetic performance, symbiont density, chlorophyll concentration, and coral polyp retraction in the short-term. Uptake of Heavy metal contaminants along with microplastic have negative effects on metabolism, fertilization, and photosynthesis of corals (Pait et al., 2007). Concurrent consumption of microplastics and HMs can alter the chromosome, causing cancer, obesity, and infertility (Islam et al., 2023; Takebira et al., 2021). In marine environments, microplastics and HMs might find their way into the food webs through the diet of corals by zooplankton and other biota, leading to their bioaccumulation and trophic transfer (Bisht & Negi, 2020). These have adverse reproductive, immunological, mutagenic, and carcinogenic effects. Hence, their impacts should be considered when assessing the impact of microplastics on marine species (Verla et al., 2019; Wang et al., 2018).

It is also important to note that the concentration of some co-absorbed chemical contaminants within microplastics can be considerably higher than the concentration in the surrounding

seawater, varying from 105 to 106 times greater (Rios et al., 2010; Ziccardi et al., 2016). This concentration also depends on how long microplastics persist in the environment. According to studies, old (black) microplastics often have greater concentrations of pollutants than colored or white microplastics (Verla et al., 2019; Wang et al., 2018). Exposure to microplastic leachate [nylon, polypropylene (PP), high-density polyethylene (HDPE), or low-density polyethylene (LDPE)] at three concentrations (50, 100, or 200 particles/L) for 7 days in Larvae of *Montipora capitata*, and the brooding species, Harbor Porites were found to affect survival and settlement of both species, the effects were more pronounced at Lechate concentration of HDPE at 100 particles/L and LDPE at 200 particles/L (Wilkins and Richmond, 2025). However, (HDPE at 200 particles/L promoted settlement unexpectedly, likely due to attractive chemical cues released by the leachates. This is particularly concerning, as it suggests that planula larvae may be drawn to settle on degraded or suboptimal substrates affecting reef recruitment and replenishment. Species-specific responses were also found, with Harbor Porites exhibiting higher survival and variable settlement. Overall, microplastic leachate significantly impacted survival and settlement of both *Montipora capitata* and Harbor Porites planula larvae with complex temporal, species-specific, polymer-type, and concentration-dependent effects (Wilkins and Richmond, 2025).

7. Effects of microplastics on Anthozoan-algae symbiotic relationships

One of the most diversified and successful marine ecosystems, coral reefs are built on the mutualistic interaction between their invertebrate hosts (scleractinian corals) and the photosynthetic dinoflagellates of the genus *Symbiodinium* (often known as zooxanthellae). This association is collectively called holobiont (van Oppen, and Burghardt, 2009). Host gastrodermal cells contain symbiosomes, which are vacuoles in which the photosynthetic algae are located

(Wakefield et al., 2000). The algal component supplies most of the energy requirements, providing about 95% of the photosynthetic products (glycerol, glucose, amino acids, or lipids) to the host (Muscatine, 1990). The symbiotic association is extremely vulnerable to environmental or anthropogenic disturbances and once disrupted, it can lead to the alarming phenomenon known as coral bleaching, which is defined by the loss of these essential symbionts. As discussed in the previous section, coral bleaching is a serious ecological concern because it has the potential to cause coral reefs to deteriorate and disappear, endangering the diverse and abundant marine ecosystems they support (Douglas, 2003).

The crucial coral-Symbiodiniaceae symbiotic relationship, that is necessary for the survival of mixotrophic corals, is significantly impacted by microplastics. According to recent studies, microplastics have the potential to interfere with this complex host-symbiont interaction between coral, and photosynthetic zooxanthellae algae (Okubo et al., 2018; Syakti et al. 2019; Okubo et al., 2020). This can occur at the initial process of infection of coral cells by the algae which is disrupted in the presence of microplastics. Okubo et al. (2018) found that ingestion of microplastics by corals hinders the establishment of the anthozoan-algae symbiotic relationship. Upon ingestion into the body of the coral, microplastics are transported from the gastrovascular cavity into the endodermal cells of the mesenterial filaments by phagocytosis, forming a phagosome membrane around the particles (Okubo et al., 2020). Since zooxanthellae use the same pathway to enter coral cells, the incorporation of microplastics into the coral endodermal cells occupies space inhabited by the symbiont thus preventing the infection of the cells by the algae. Smaller-sized microplastics remain in the cells for a longer time compared to particles with large sizes (Okubo et al., 2018)

Along with the immediate impacts, microplastics can also cause secondary stressors and ROS production (Mendrik et al., 2021) in the symbiotic interaction. The increased oxidative stress

levels, can harm cellular components and interfere with their regular physiological functions. For example, it causes tissue necrosis and apoptosis, which may have an impact on the stability of the coral-algae relationship (de Oliveira Soares et al., 2020). Chlorophyll content in the coral symbiont decreases in the presence of microplastics, which is linked to the interaction of microplastics and the associated toxic compounds and microbes within coral tissues and the overproduction of ROS which impairs coral's essential photosystem II ability to function. Also, microplastic attachment to the coral skeleton decreases the passage of light energy, which reduces the photosynthetic efficiency of zooxanthellae (Xiao et al., 2021). In addition, microplastics may cause a reduction in nutrient uptake within the endosymbiont, a crucial function for both coral and algae, potentially resulting in nutritional deficiency that weakens the coral. They are able to reduce the number and density of these algal cells. The equilibrium of the symbiotic interaction is interfered with, and as a result of the decreased algal growth, the photosynthetic efficiency of the symbiont reduces, hence corals receive less nourishment and less energy, which may impair their overall health and resilience (de Oliveira Soares et al., 2020). Moreover, decreased algal growth inside coral tissues is an important factor in causing coral bleaching (Douglas, 2003).

Figure 3 shows the effects of microplastics on coral ecosystems (coral bleaching, tissue necrosis, and growth reduction).

8. Trophic level transfer of microplastic in coral ecosystems

As coral reefs play a crucial role in the start of the food chain in the marine ecosystem, corals and the associated algae gradually affect the life of organisms at higher trophic levels as a result of exposure to cumulative contaminants (Tebbett and Bellwood, 2021). In coral reef ecosystems, harmful effects and other related disorders can be brought on by microplastics as well as pathogenic microbes and chemicals that are carried together with microplastic particles.

Microplastics can enter the cells of the phytoplankton (producers) by diffusion across the cell walls due to their small size. In this way, these microplastic particles enter the food chain and are transferred to higher trophic levels, a process called ‘Trophic transfer’ (Bisht et al., 2020). Primary consumers (zooplankton) ingest microplastics either directly or indirectly by feeding on phytoplankton. A study by Cole et al. (2013) on microplastic ingestion by zooplanktons shows thirteen zooplankton taxa (including copepods, bivalve larvae, and decapod larvae) could ingest microplastics in a range of 1.7–30.6 μm of polystyrene beads. Moreover, zooplankton can retain microplastics in their system for a long period (Kaposi et al., 2014), thus resulting in the trophic transfer of microplastic fragments to coral organisms (secondary consumers). When corals feed heterotrophically on these zooplankton, they indirectly take up the microplastics accumulated in the plankton (Hankins et al., 2018). Therefore, these indigestible particles get incorporated into the tissues of corals after direct ingestion of the particles (Hall et al., 2015) or indirectly through feeding on zooplanktons that also ingest microplastics (Ferrier-Pagès et al. 2003). Upon ingestion, the microplastic particles enter the gastrovascular cavity and then are phagocytosed into the endodermal tissues (Okubo et al., 2020). Marine worms, fish species, barnacles, snails, and other organisms that feed on coral tissues and the associated algal species are also exposed to the toxic effects of these contaminants. They are subject to bio-accumulative effects (John et al., 2022).

Figure 4 shows different routes of trophic transfer of microplastics in coral ecosystems.

9. Policy options for microplastic mitigation in coral ecosystems

A comprehensive and coherent strategy integrating research, innovation, and evidence-based policy is essential to mitigate the increasing threat of microplastics to coral reef ecosystems. While current efforts emphasize reducing plastic waste at its source, strengthening waste management, and promoting biodegradable alternatives (Spring et al., 2025), important research gaps remain.

For instance, the efficiency of community-driven clean-up programmes in reducing microplastic flux into coral reefs is still poorly quantified, and long-term monitoring frameworks to assess their ecological benefits are largely absent. Moreover, although strict regulations on microplastic emissions are frequently recommended, the effectiveness of enforcement mechanisms, especially in low- and middle-income coastal nations, requires systematic study. Interdisciplinary collaboration among marine ecologists, environmental scientists, policymakers, industries, and local communities will therefore be crucial to test and refine these interventions.

A major upstream challenge is the release of microplastics from textiles, yet research on emission rates under real-world washing conditions, the environmental persistence of synthetic fibres, and the efficacy of advanced washing machine filters remains limited. Technological innovations such as closed-loop polymer capture systems, microfiber filters, and biodegradable fibre alternatives are promising, but require robust life-cycle assessments to evaluate their ecotoxicity, cost-effectiveness, and scalability. Similarly, although shoreline clean-up operations can reduce large plastic debris that eventually fragments into microplastics, studies quantifying their contribution to lowering microplastic loads in coral reef environments are sparse and should be prioritized.

The transition to biodegradable polymers (e.g., PHAs, PLAs, PHBs) represents another emerging avenue, yet critical gaps persist regarding their degradation pathways in marine environments, potential by-products, and impacts on coral-associated microbiomes. Their production from waste materials such as sewage sludge offers circular-economy benefits (Ogunola et al., 2018), but the ecological safety and feasibility of deploying such biopolymers in high-risk sectors like fisheries demand more targeted evaluation.

At the global policy level, the failure of the plastics treaty negotiations at INC-5.2 in Geneva (August 2025) underscores major governance gaps, particularly concerning production caps, limits

on chemicals of concern, and binding global controls (Farrelly et al., 2025; Knoblauch & Mederake, 2024). Research is urgently needed to understand how political–economic interests of petrochemical-exporting nations shape international environmental agreements. The absence of a unified treaty leaves escalating risks unaddressed, especially as microplastics are increasingly detected in human tissues, highlighting a growing public health emergency (University of Portsmouth, 2025). Although national policies such as the US Microbead-Free Waters Act of 2015 and similar bans in the UK and Canada have reduced microbead contamination, enforcement gaps persist—particularly for emerging synthetic substitutes such as microplastic waxes—indicating a pressing need for regulatory expansion and improved compliance monitoring.

The establishment of the Intergovernmental Science-Policy Panel on Chemicals, Waste and Pollution (ISP-CWP) by UNEP offers a promising platform to integrate scientific evidence into policy design (UNEP, 2025). The first session of the Plenary (ISP-CWP P1) was hosted by the Government of Switzerland from February 2-6 2026, in Geneva. Hopefully ISP-CWP will be able to influence national decision-making, harmonizes monitoring standards, and supports mitigation strategies tailored to diverse socioeconomic contexts. The Panel’s upcoming initiatives can be effective in managing microplastic pollution and toxicity in the marine ecosystems, including the coral ecosystems.

Finally, plastic production is closely linked to greenhouse gas emissions, and thus to global warming and ocean acidification—both major drivers of coral decline. However, research connecting plastic life-cycle emissions with localized coral reef impacts remains limited. Integrating climate mitigation strategies with plastic pollution reduction could improve both the ecological and physiological resilience of coral communities, yet frameworks that operationalize this joint approach are still lacking.

Overall, despite significant progress, critical research gaps remain in understanding microplastic sources, emission pathways, degradation dynamics, ecological impacts on corals, and the effectiveness of policy interventions. Addressing these gaps through interdisciplinary, multi-scalar research, coupled with improved international cooperation, will be essential for developing comprehensive, science-based strategies to protect coral reef ecosystems and ensure the long-term resilience of marine biodiversity.

10. Conclusions

This paper demonstrates and explains multiple physiological, biochemical, and ecological threats posed by microplastic pollution to coral reef ecosystems. The evidences showed that microplastics disrupt coral feeding behaviour, induce oxidative stress, impair growth and reproduction, facilitate pathogen transmission, and destabilize the coral–algae symbiosis essential for reef functioning. These impacts cascade across trophic levels, altering reef food webs and ultimately affecting human populations through contaminated seafood consumption. Although research on microplastic–coral interactions has expanded rapidly, several critical knowledge gaps hinder comprehensive risk assessment and the development of effective mitigation strategies.

Deep-sea and cold-water corals remain particularly understudied, even though their responses to microplastic exposure may differ markedly from those of shallow-water species. Expanding research into these ecosystems is essential for identifying species- and habitat-specific vulnerabilities. Additionally, long-term ecological studies are scarce; most existing work focuses on short-term laboratory experiments that do not capture chronic or multi-generational impacts. Field-based, long-duration investigations are urgently needed to evaluate cumulative effects under realistic environmental conditions and determine whether coral populations can acclimate or adapt

to persistent microplastic pollution, or whether they face long-term genetic and demographic decline.

Species-specific variation in microplastic sensitivity is another major gap. Although studies indicate considerable differences among coral taxa, the biochemical and molecular bases for these disparities remain largely unknown. Comparative transcriptomic and proteomic analyses would help clarify underlying mechanisms and identify reliable stress biomarkers. Furthermore, interactions between microplastics and other environmental stressors, such as ocean warming, acidification, eutrophication, and disease, are poorly understood. Realistic multi-factorial experiments are required to predict ecosystem responses under climate-driven conditions and to identify potential tipping points.

Trophic transfer and biomagnification of microplastics through reef food webs, culminating in human exposure, also remain inadequately quantified. Field studies tracing microplastic movement across multiple trophic levels are essential for accurate human health risk assessment. Similarly, the role of microplastics as vectors for pollutants and pathogens demands intensified research. Advanced analytical approaches are needed to characterize contaminant desorption kinetics within coral tissues and to distinguish microplastic-associated contaminant exposure from direct environmental sources.

Finally, mitigation strategies—including biodegradable polymer alternatives, microplastic removal technologies, and regulatory interventions—require rigorous field validation. Their feasibility, efficacy, and potential unintended ecological consequences must be critically evaluated before large-scale implementation. Addressing these research gaps will require interdisciplinary collaboration, standardized methodological frameworks, and strong international research networks. Coupling long-term monitoring programmes with adaptive management will accelerate

knowledge generation and support the translation of scientific evidence into effective policy decisions for the conservation and resilience of coral reef ecosystems.

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Figure captions:

Figure 1: Global abundance of microplastics in coral reef ecosystems. The light blue polygon represents the sampled locations, within and outside LME's. Map acquired using GIS. Reprinted from Elsevier (Slynkova et al., 2025), open access, creative commons license 4.0.

Figure 2: Attachment and subsequent ingestion of microplastics by corals, leading to physiological effects on corals.

Figure 3: Effects of microplastics on coral ecosystems, such as coral bleaching, tissue necrosis, growth reduction.

Figure 4: Different routes of trophic transfer of microplastics in coral ecosystems.

		PE: 25 MPs/L, 100 MPs/L, 200 MPs/L	48 h	effect was not concentration dependent No significant effect on embryo development and larval settlement	
<i>Zoanthus sociatus</i>	Aged PE fragments (63–125 µm), PVC fragments (63–125 µm)	1 mg/L (PE: ~1040–8125 MPs/L; PVC: ~708–5534 MPs/L*) 10 mg/L (PE: ~10,402–81,255 MPs/L, PVC: ~7085–55,347 MPs/L*)	96 h	microplastic -specific impact observed PVC (10 mg/L) increased photosynthetic efficiency, lipid peroxidation and antioxidant defenses	Rocha et al. (2020)
<i>Porites porites</i>	PS pellets (<20 µm)	1 mg/L (~1.5 ×10 ⁷ MPs/L*) 10 mg/L (~1.5 ×10 ⁸ MPs/L*)	96 h	No significant difference in catalase enzymatic activity or melanin content for all concentrations	Grillo et al. (2021)

				ATP, and concentrations of TAC and GSH. PET microplastics had the most significant impact	
<i>Acropora</i> sp.	PE fragments, PET fragments, PA 66 fragments	50 mg/L	96 h	Microplastics caused decrease in endosymbiont density and chlorophyll content	Xiao et al. (2021)
<i>Gonioporacolumna</i>	PE powder (40–48 µm)	5 mg/L (~1.25 ×10 ⁵ MPs/L [*]) 10 mg/L (~2.5 ×10 ⁵ MPs/L [*]) 50 mg/L (~1.3 ×10 ⁶ MPs/L [*]) 100 mg/L (~2.5 ×10 ⁶ MPs/L [*])	7 days	Symbiont density significantly increased at different microplastics concentrations compare to control At different microplastics concentrations catalase, glutathione peroxidase, glutathione transferase, glutathione, malondialdehyde activity was increased	Chen et al. (2022)

		50 mg/L (~1.3 ×10 ⁴ MPs/L*) 70 mg/L (~1.8 ×10 ⁴ MPs/L*)		Concentration above 50 mg/L caused significantly different cellular response to corals exposed to 10 mg/L	(2022)
<i>Tubastrea aurea</i>	PVC beads (5–10 µm)	1 mg/L (~1.1 ×10 ⁷ –1.4 ×10 ⁶ *) 30 mg/L (~4.2 ×10 ⁷ –3.3 ×10 ⁸ *)	72 h	Co-exposure of microplastics and Cu(II) caused significant decrease in lipid peroxide activity Microplastics decreased the toxicity of Cu(II)	Xiao et al. (2022)
<i>Pinnigorgia flava</i>	Artificially aged PP fibers	0.001 mg/L 0.1 mg/L	72 h	Microplastics did not cause mortality Concentration above 1 mg/L caused the	Isa et al. (2023)

				Overall, impact of microplastics was less significant to increase temperature	
<i>Briareum violacea</i>	PE fragments (40–48 µm)	5 mg/L (~1.25 ×10 ⁵ MPs/L [*]) 10 mg/L (~2.5 ×10 ⁵ MPs/L [*]) 50 mg/L (~1.3 ×10 ⁶ MPs/L [*]) 100 mg/L (~2.5 ×10 ⁶ MPs/L [*]) 300 mg/L (~7.5 ×10 ⁶ MPs/L [*])	7 days	Microplastics caused skeleton degradation — 10 mg/L microplastics caused transformation of coral skeletons from aragonite calcium carbonate crystals to amorphous calcium carbonate	Lin et al. (2025)

Coral Species	MP Type	MP concentration	Exposure Duration	Observation	Reference
<i>Lopheliapertusa</i>	Biofouled LDPE beads (500 µm)	350 MPs/L	2 months	Corals growth was significantly reduced Microplastics did not affect polyp activity, prey capture rate, but calcification was decreased	Chapron et al. (2018)
<i>Acropora humilis</i> , <i>Acropora millepora</i> , <i>Pocillopora verrucosa</i> , <i>Pocilloporadamicornis</i> , <i>Porites lutea</i> , <i>Porites cylindrica</i>	PE fragments (37–163 µm) (biofilm formed during exposure)	3799 ±546 MPs/L	28 d	Species-specific impact with <i>P. verrucosa</i> being affected the most Microplastics cause negative effects on health such as bleaching and tissue necrosis	Reichert et al. (2018)
<i>Montipora capitata</i> , <i>Pocilloporadamicornis</i>	PE beads (150–180 µm)	2000 MPs/L	10 d with 1 h feeding	Species-specific effect observed	Axworthy and Padilla-

<i>Acropora formosa</i>	PE fragments (<100 µm, 100–200 µm and 200–500 µm)	50 mg/L (~812–1 ×105 MPs/L*) 100 mg/L (~1625–2 ×105 MPs/L*) 150 mg/L (~2438–3 ×105 MPs/L*)	14 days	photosynthetic efficiency Microplastics cause dose and size specific effect on corals health. Fragments <100 µm at 150 mg/L caused the most severe impact with 93.6 % corals experiencing bleaching and 5.9 % tissue necrosis	Syakti et al. (2019)
<i>Stylophorapistillata</i>	PE beads (106–125 µm)	5000 MPs/L 50,000 MPs/L	28 d	Photosynthetic activity was impacted at higher concentration Higher concentration caused significant change to metabolite profiles	Lancôt et al. (2020)
<i>Pseudodiploriaclivosa</i> , <i>Acropora cervicornis</i>	PE fragment (212–250 µm, 425–500 µm, 850–1000 µm, 1700–2000 µm)	30 mg/L (~2.8–6397 MPs/ L*)	3 months	Species-specific effect observed MPs significantly decreased	Hankins et al. (2021)

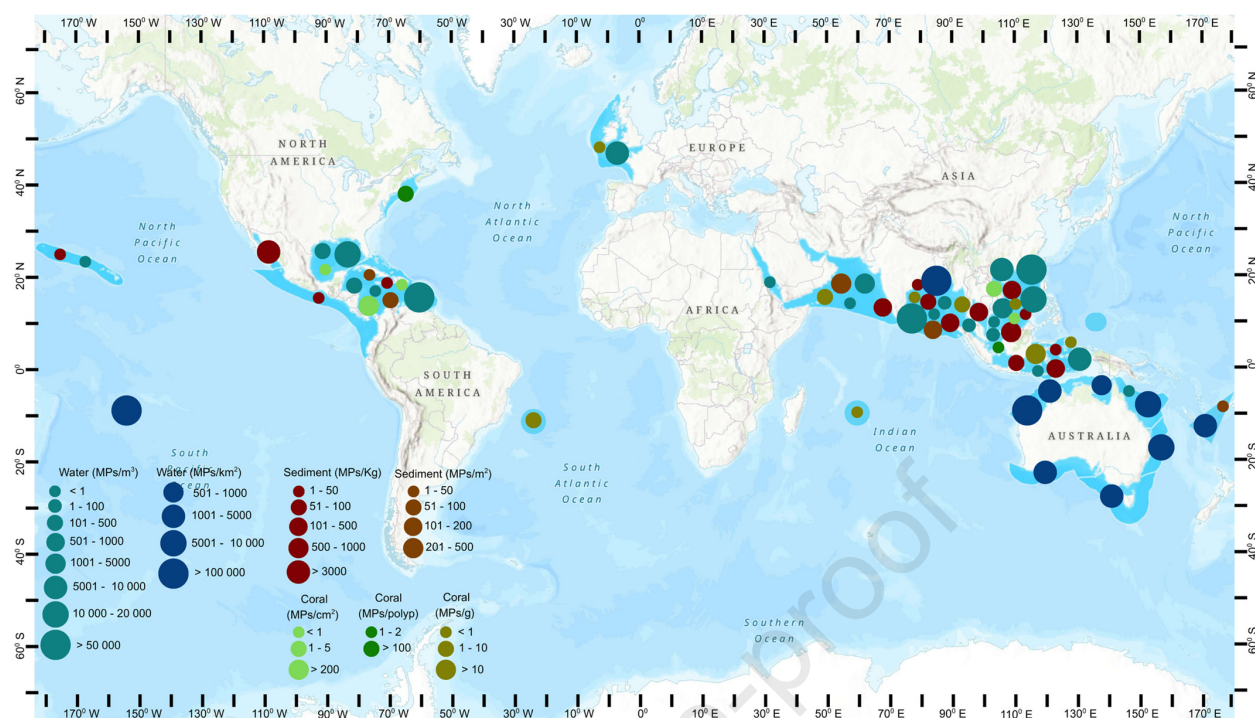
				and oxidation resistance were also impacted by microplastics	
<i>Acropora</i> sp., <i>Seriatopora hystrix</i>	PS beads (500–1000 µm), fibers of unknown polymer (500 µm–1 cm)	0.1 mg	12 d	Microplastic-specific and species-specific effect observed. Fibers significantly reduced photosynthetic activity of <i>Acropora</i> sp. Beads significantly increased photosynthetic activity of <i>S. hystrix</i>	Mendrik et al. (2021)
<i>Acropora muricata</i> , <i>Pocillopora verrucosa</i> , <i>Porites lutea</i> , <i>Helioporacoerulea</i>	PE fragments (65–200 µm) (Biofilm formed over the course of exposure)	2.5 mg/L (~634–18,495 MPs/L*)	75 week	Low number of microplastics accumulated in coral tissues (up to 2 particles/cm ²), up to 84 particles/cm ² found in coral skeleton	Reichert et al. (2022)

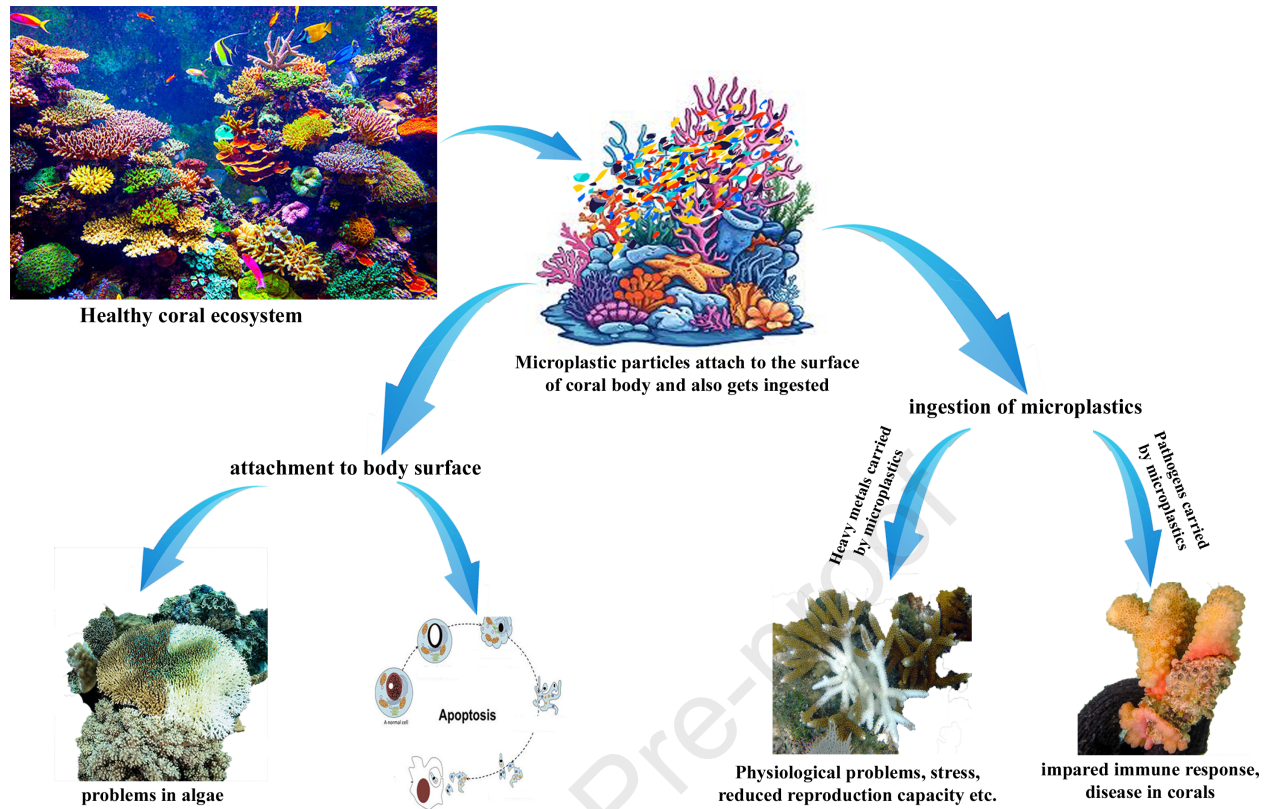
		10 mg/L (~2.5 ×10 ⁵ MPs/ L [*]) 50 mg/L (~1.3 ×10 ⁶ MPs/ L [*]) 100 mg/L (~2.5 ×10 ⁶ MPs/L [*]) 300 mg/L (~7.5 ×10 ⁶ MPs/L [*])		communities 300 mg/L caused increase in the abundance of <i>Donghicola</i>	
<i>Acropora cervicornis</i>	Biofouled PE beads (212–355 µm)	11.8 MPs/L	17 d	No significant effect (bleaching, tissue necrosis and zooxanthellae density) of microplastics on <i>A. cervicornis</i>	Plafcan and Stallings (2022)
<i>Acropora muricata</i> , <i>Pocillopora verrucosa</i> ,	Biofouled PE fragments (184 µm)	200 MPs/L	15 months with 24 h	Microplastics did not cause changes in defense mechanisms	Rades et al. (2022)

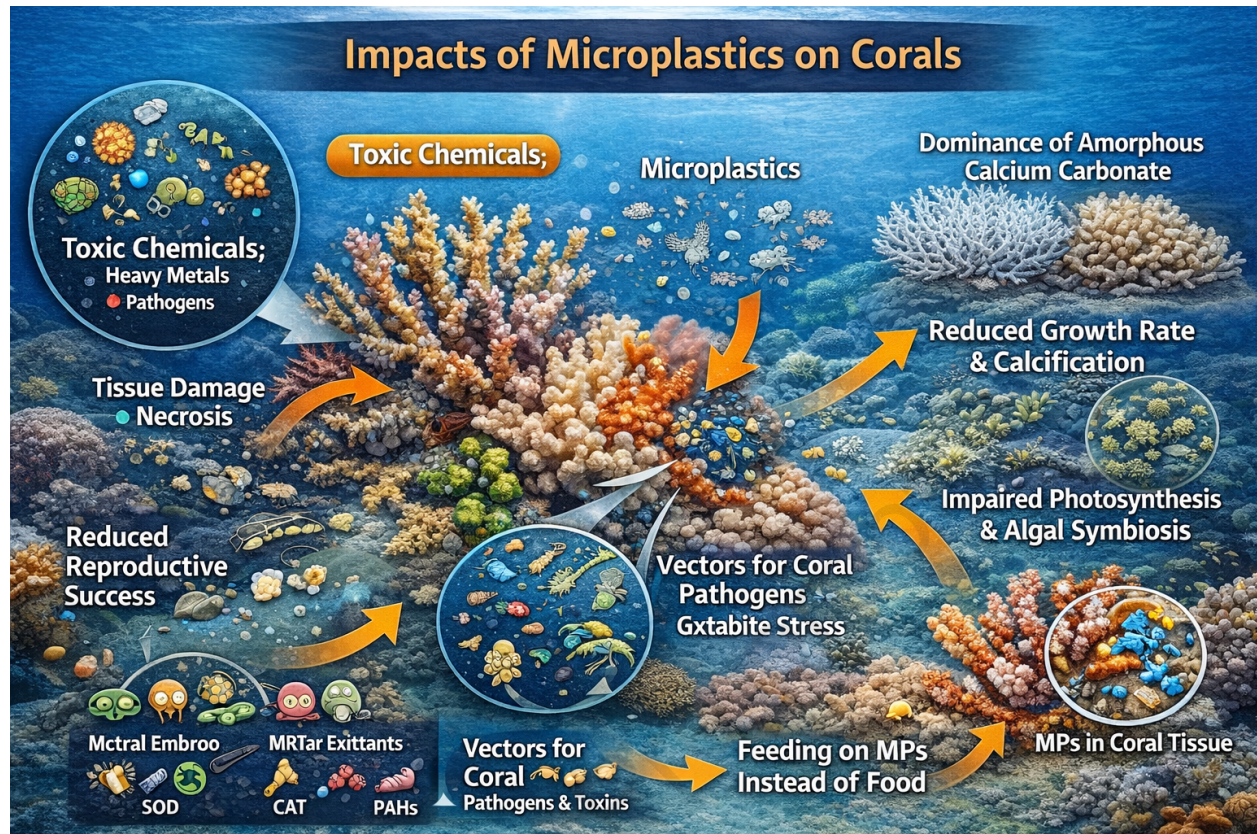
	PEI and other) (100–1000 µm), MPs fibers (acrylic, PA, PES) (1006 µm), tire wear (100–355 µm)			<i>verrucosa</i> more significantly than <i>S. pistitata</i> Microplastics caused decreased calcification in <i>P. verrucosa</i> Photosynthetic efficiency increased in both species	
<i>Turbinariamesenterina</i>	PS beads (9.5–11.5 µm)	0.01 mg/L (~11,959–21,214 MPs/L*) 0.1 mg/L (~1.2 ×10⁵–2.1 ×10⁵ MPs/L*)	10 d	Microplastics cause no significant effect on symbiotic algae density, photosynthetic efficiency or chlorophyll content Oxidative stress significantly increase in both MPs concentrations	Tang et al. (2024)
<i>Pocilloporadamicornis</i> ,	Pristine and aged PS	0.1 mg/L (~22–249	14 d	Species-specific impact observed	Washington

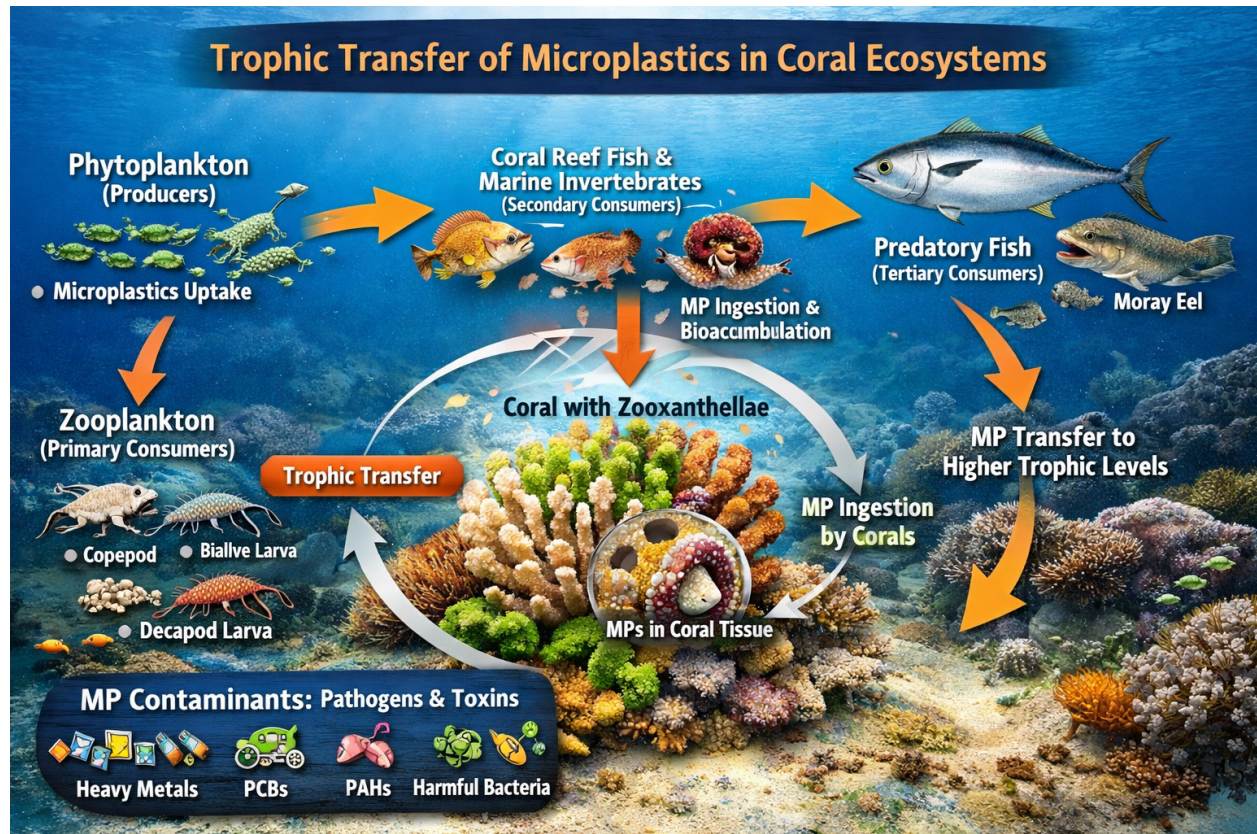
Great Barrier Reef, Australia	Water	0.355–5 mm	0.04–0.46 MP m ⁻³	Jensen et al., 2019
Australian waters	Water	0.4–82.6 mm	$4.26 \times 10^{-2} \pm 7.4 \times 10^{-3}$ MP km ⁻² (0–4.89 $\times 10^{-2}$ MP m ⁻²)	Reisser et al., 2013
Mo'orea, French Polynesia	Water	0.05–5 mm	0.74 MP m ⁻²	Connors, 2017
Faafu Atoll, Maldives	Water	330 μm –5 mm	0.26 particles m ⁻³ (0.02–0.48 MP m ⁻³)	Saliu et al., 2018
	Water	200 μm –5 mm	0.12 ± 0.09 particles m ⁻³ (0.03–0.65 MP m ⁻³)	Saliu et al., 2019
Xisha Islands, South China Sea	Water	7–4856 μm	200–45,200 MP m ⁻³	Ding et al., 2019

Vavvaru Island, Maldives	Sediment	>1 mm	35.8 ± 42.5 MP m ⁻² ; 1029 ± 1134 MP m ⁻² in accumulation zones	Imhofet al., 2017
Four Coral Islands, Hong Kong	Sediment	0.3–5 mm	194.5 ± 49.9 MP kg ⁻¹ (171.7 ± 57.6 to 223 ± 51.4 MP kg ⁻¹)	Cheang et al., 2018
Xisha Is. Nansha is, Weizhou Is. and Sanya Lu Hui Tou, South China Sea	Sediment	0.3–5mm	60 ± 3 to 610 ± 11MP kg ⁻¹ ; 40 ± 4 to 100 ± 2 MP kg ⁻¹ ; 60 ± 2 to 90 ± 5 MP kg ⁻¹ , and; 50 ± 3 to 350 ± 7 MP kg ⁻¹ , respectively	Zhang et al., 2019
Gulf of Mannar, India	Sediment	0.8–5 mm	50–103.8 MP kg ⁻¹	Patterson et al., 2020
	Sediment	2–5 mm	55 ± 2a1 to 259 ± 88 MP kg ⁻¹	Jeyasanta et al.,2020
Lhaviyani Atoll, Maldives	Sediment	63 μ m–4 mm	277.90 ± 24.98 MP kg ⁻¹ (55–1127.5 MP kg ⁻¹)	Patti et al.,









Highlights:

- Microplastic pollution leads to reduced food intake and energy depletion in corals.
- Microplastics induce oxidative stress in coral tissues, leading to diseases.
- Microplastics also act as vectors for contaminants, affecting corals.
- Anthozoan-algae symbiotic relationship is affected by microplastic pollution
- International cooperation is required to mitigate microplastic pollution in coral ecosystems.

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: